Evaluating Preference Adaptation via Post-Interaction Decision Consistency: A Theoretical and Experimental Framework from Simulated Preference Agents to Real Humans

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Abstract

Interactive personalization for large language model (LLM) agents is increasingly studied in simulated environments, yet recent evidence suggests that LLM-based user simulators are often unreliable proxies for real users and exhibit substantial Sim2Real gaps. Existing work therefore faces a dual limitation: it either over-relies on simulated users for evaluation and optimization, or it studies personalization without a formal account of what exactly is transferred from simulation to real human interaction. In this paper, we propose a different perspective. We argue that simulation should not be treated as a surrogate for human preference itself, but as a controlled task family for validating whether an agent's *interactive learning mechanism* can fit decision-relevant preference patterns from behavioral data. Motivated by meta-learning, we model each preference type as a task, train the agent across a diverse family of simulated preference tasks, and evaluate whether the learned adaptation rule can quickly fit unseen preference-induced decision behavior from interaction trajectories. We formalize this setting, derive a transfer guarantee under approximate task-family alignment and smoothness of adaptation risk in task space, and propose a practical experimental protocol spanning simulated multi-task validation and real-human deployment. Our framework yields a precise and testable thesis: if an agent's interactive learning mechanism can fit diverse preference-induced decision patterns in a broad, controlled task family, then when the interaction partner is replaced by a real human whose behavior is drawn from a nearby task family, the same mechanism should enable the agent to fit the human's future decision behavior through direct interaction.

1 Introduction

Personalized agents must infer user preferences from interaction and adapt future decisions accordingly. This requirement appears in tool-augmented assistants, recommender agents, planning agents, and writing assistants. However, current evaluation and training pipelines often rely on simulated users because large-scale human evaluation is expensive and operationally difficult. Recent work has shown that such reliance is problematic: LLM-simulated users can be unreliable proxies for real users, with evaluation outcomes varying across simulator choices and failing to faithfully reflect real interaction dynamics [Seshadri et al., 2026]. More recently, large-scale human studies demonstrated a substantial Sim2Real gap: simulated users tend to be excessively cooperative, stylistically homogeneous, and unrealistically positive, thereby inflating agent performance and masking real-world failure modes [Zhou et al., 2026].

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These findings reveal a central weakness in the current literature. If the simulator is not a faithful proxy, then high performance against the simulator does not imply that the agent has learned real human preferences. At the same time, abandoning simulation altogether is undesirable, because simulation provides a scalable and controlled mechanism for generating diverse preference conditions and repeated adaptation episodes. This tension motivates the central question of this paper:

> *What, exactly, can be validated in simulation, and under what conditions does such validation support direct human preference fitting?*

Our answer is to shift the object of validation. Instead of treating simulation as a substitute for human preference, we treat simulation as a *task distribution* for validating whether the agent's *interactive learning mechanism* can fit decision-relevant preference patterns from behavior. This viewpoint is inspired by meta-learning [Finn et al., 2017]: rather than optimizing the agent for one fixed user profile, we expose it to a family of preference tasks and explicitly train it to become easy to adapt from sparse interaction data. Under this view, each preference type defines a task; interaction trajectories provide support examples; and future preference-consistent decisions serve as query examples. Success in simulation then means not that the agent has imitated the simulator, but that the agent has learned a transferable adaptation procedure for fitting future behavior induced by latent preferences.

Based on this perspective, we formulate the following paper thesis.

Paper Thesis. *If an agent's interactive learning mechanism can fit diverse preference-induced decision patterns from behavioral data across a rich family of simulated preference tasks, then when the interaction partner is replaced by a real human—and provided the human preference dynamics are drawn from a nearby task family—the same mechanism should enable the agent to fit the human's future decision behavior through direct interaction.*

This thesis is weaker, more precise, and more defensible than the common but flawed claim that success against simulated users directly implies success on humans. It explicitly separates two notions:

1. **Simulator fidelity:** whether the simulator itself resembles real users.
2. **Adaptation mechanism validity:** whether the agent can fit decision-relevant preference patterns from interaction data.

Our framework focuses on the second. Simulation is used to validate the adaptation mechanism across a controlled family of preference structures; real-human interaction is then used to instantiate the same mechanism directly on humans, without requiring the simulator to serve as a perfect human proxy.

Contributions. This paper makes four contributions:

1. We formalize interactive preference adaptation as a meta-learning problem over latent preference tasks.
2. We introduce a mathematical framework that distinguishes simulator imitation from preference-pattern fitting, and derive a transfer bound under weak task-family alignment assumptions.
3. We propose a principled metric suite centered on decision similarity, ranking consistency, adaptation efficiency, and simulated distributional fit, while clarifying why raw simulator agreement alone is insufficient.
4. We provide a complete experimental blueprint, including simulated task-family validation, ablations on task-family coverage, and real-human deployment experiments that directly test whether the learned mechanism fits human future decisions.

2 Related Work

LLM user simulation and its limitations. A growing body of work uses LLMs as user simulators for interactive agent evaluation and optimization. However, recent evidence questions the validity of this paradigm. Seshadri et al. [2026] show that LLM-simulated users are unreliable proxies for real

users in agentic evaluations, with substantial variability across simulator models and demographic discrepancies. Zhou et al. [2026] systematically characterize the Sim2Real gap in user simulation, showing that simulated users are overly cooperative, lack realistic frustration and ambiguity, and produce unrealistically positive feedback. These findings motivate our central design choice: simulation should not be interpreted as a direct replacement for real users, but rather as a controlled environment for validating the agent's adaptation mechanism.

Personalized agent evaluation. Recent personalized agent benchmarks evaluate preference-following, personalization quality, and tool invocation under user-specific profiles. ETAPP, for example, studies personalized tool-augmented LLMs and emphasizes that personalization must be evaluated jointly with correct task execution and tool use [Hao et al., 2025]. These benchmarks are important but do not offer a formal account of when adaptation performance in simulated settings should translate to real human interaction. Our work complements this line by providing both a theory and an evaluation protocol for testing whether interactive preference learning mechanisms generalize beyond simulation.

Meta-learning and rapid adaptation. Meta-learning aims to acquire parameters or update rules that are easy to adapt using few examples from new tasks. Model-Agnostic Meta-Learning (MAML) is a canonical instance, optimizing initial parameters such that a small number of gradient steps on a new task leads to good generalization [Finn et al., 2017]. Our work is conceptually aligned with this principle, but the tasks are not image classes or regression functions; instead, they are latent user preference structures expressed through interaction trajectories. The meta-learning perspective is crucial because it reframes simulation from a proxy-evaluation device into a task distribution for validating the transferability of the adaptation mechanism.

3 Problem Formulation

We model an interactive decision problem as follows.

Environment and decision space. Let $x \in \mathcal{X}$ denote a decision context (query, situation, state, or task instance), and let $y \in \mathcal{Y}$ denote a decision (action, ranked choice, tool invocation, recommendation, or structured response).

Latent preference structure. Each user instance is associated with a latent preference parameter $\theta \in \Theta$. The user induces a conditional decision distribution

$$P_\theta(y \mid x). \tag{1}$$

This distribution captures how a user with latent preference θ resolves trade-offs in context x.

Interaction history. The agent interacts with a user over rounds and observes an interaction trajectory

$$\mathcal{D}_t = \{(x_1, y_1), (x_2, y_2), \ldots, (x_t, y_t)\}, \tag{2}$$

where y_i is the user's decision or preference-bearing feedback at round i.

Adaptive agent. Agent B maintains an adaptation state $z_t = \mathcal{A}_\phi(\mathcal{D}_t)$, where $\mathcal{A}_\phi$ is an interaction-conditioned learning mechanism parameterized by ϕ. Conditioned on z_t, the agent predicts future decisions through

$$Q_\phi(y \mid x, \mathcal{D}_t). \tag{3}$$

The goal is not simply to produce high reward, but to fit enough of the user's decision-relevant preference structure so that the agent's future decisions are aligned with the user's decision distribution.

Task-family view. A *task* is defined by a latent preference θ and an associated context distribution $\mu_\theta(x)$. A task family is thus

$$\mathcal{T} = \{(\mu_\theta, P_\theta) : \theta \in \Theta\}. \tag{4}$$

Simulation provides a scalable way to instantiate many tasks from $\mathcal{T}$.

4 Core Thesis and Theoretical Framework

4.1 Why simulator agreement alone is insufficient

Suppose A is a simulator and B is an adaptive agent trained or tested against A. A common evaluation metric is decision agreement:

$$\mathrm{DA}(A, B \mid \theta) = \mathbb{E}_{x\sim\mu_\theta}\left[\mathbf{1}\big(f_A(x,\theta) = f_B(x,\mathcal{D}_t)\big)\right], \tag{5}$$

where f_A and f_B denote deterministic top-1 decisions.

A high value of $\mathrm{DA}(A, B \mid \theta)$ only establishes that B agrees with the simulator-defined decision rule. This does *not* imply that B has learned real human preferences unless the simulator is independently known to match real users. Recent work shows that such an assumption is often false [Seshadri et al., 2026, Zhou et al., 2026].

Hence we reinterpret simulated agreement as evidence about the *adaptation mechanism*, not evidence that the agent has already fit human preferences.

4.2 Interactive preference fitting as meta-learning

We define the support set as the interaction history $\mathcal{D}_t$ and the query set as held-out contexts $x \sim \mu_\theta$ on which the agent must predict or match the user's future decisions. This induces a meta-learning objective:

$$\min_{\phi}\ \mathbb{E}_{\theta\sim\Pi_{\mathrm{sim}}}\left[\mathbb{E}_{\mathcal{D}_t\sim P_\theta}\mathbb{E}_{x\sim\mu_\theta}\ell\big(Q_\phi(\cdot \mid x,\mathcal{D}_t), P_\theta(\cdot \mid x)\big)\right], \tag{6}$$

where ℓ is a proper scoring or divergence loss.

This objective explicitly optimizes the agent to adapt from sparse behavior data across many preference tasks, mirroring the meta-learning principle of being easy to fine-tune on unseen tasks [Finn et al., 2017].

4.3 Preference recovery risk

Definition 1 (Preference recovery risk)**.** *For a task θ, define*

$$R_t(\phi;\theta) = \mathbb{E}_{x\sim\mu_\theta}\left[d\big(Q_\phi(\cdot \mid x,\mathcal{D}_t), P_\theta(\cdot \mid x)\big)\right], \tag{7}$$

where d is a metric or divergence over decision distributions (e.g., total variation, Jensen–Shannon divergence, or cross-entropy-induced regret).

We write d_Θ for a task-space metric over preference tasks. Depending on the application, d_Θ may be defined by Euclidean distance between latent preference vectors, by a distance between task-generating utilities, or by a distance induced by the corresponding decision distributions.

4.4 Assumptions

Assumption 1 (Coupled approximate coverage)**.** *There exists a coupling γ between the human task distribution Π_{human} and the simulated task distribution Π_{sim} such that*

$$(\theta_h,\theta_s)\sim\gamma \quad\Longrightarrow\quad d_\Theta(\theta_h,\theta_s)\le\delta \quad \textit{almost surely},$$

for some $\delta \ge 0$.

Assumption 2 (Risk smoothness)**.** *For a fixed interaction length t and learned adaptive mechanism ϕ, the preference recovery risk is Lipschitz in task space: there exists $L_t > 0$ such that for all $\theta, \theta' \in \Theta$,*

$$|R_t(\phi;\theta) - R_t(\phi;\theta')| \le L_t\, d_\Theta(\theta,\theta').$$

4.5 Main theorem

Theorem 1 (From simulated task-family adaptation to direct human preference fitting)**.** *Suppose an adaptive agent B with parameters ϕ is meta-trained over a simulated task distribution Π_{sim} and satisfies*

$$\mathbb{E}_{\theta\sim\Pi_{\mathrm{sim}}} R_t(\phi;\theta) \le \varepsilon_t. \tag{8}$$

Under coupled approximate coverage and risk smoothness,

$$\mathbb{E}_{\theta\sim\Pi_{\text{human}}} R_t(\phi;\theta) \le \varepsilon_t + L_t\delta. \tag{9}$$

In particular, if the simulated adaptation risk is small and the human task family is close to the simulated task family in task space, then the same mechanism achieves low human preference-fitting risk under direct human interaction.

Proof sketch. Let γ be the coupling from Assumption 1. Then

$$\mathbb{E}_{\theta\sim\Pi_{\text{human}}} R_t(\phi;\theta) = \mathbb{E}_{(\theta_h,\theta_s)\sim\gamma} R_t(\phi;\theta_h).$$

By Assumption 2 and the fact that $d_\Theta(\theta_h,\theta_s) \le \delta$ almost surely under γ,

$$R_t(\phi;\theta_h) \le R_t(\phi;\theta_s) + L_t\delta.$$

Taking expectation with respect to γ gives

$$\mathbb{E}_{\theta\sim\Pi_{\text{human}}} R_t(\phi;\theta) \le \mathbb{E}_{\theta\sim\Pi_{\text{sim}}} R_t(\phi;\theta) + L_t\delta.$$

Applying the simulated-risk premise yields the stated bound. □

Interpretation. The theorem formalizes the exact claim we want. The point of simulation is not to prove that the simulator equals the human. Rather, simulation validates that the adaptation mechanism achieves low fitting risk over a broad task family. If the human task distribution is close to the simulated task distribution in task space, and if the induced risk varies smoothly across nearby tasks, then low simulated risk transfers to low human fitting risk.

4.6 Decision similarity as an observable proxy

For evaluation, the simplest observable metric is decision similarity:

$$\mathrm{DS}_t(\theta) = \mathbb{E}_{x\sim\mu_\theta}\left[\mathrm{sim}\big(f_B(x,\mathcal{D}_t), f_\theta(x)\big)\right], \tag{10}$$

where sim may be exact-match accuracy, top-k overlap, or rank correlation. This metric is useful but insufficient on its own, because two models can agree on top-1 decisions while differing substantially in uncertainty or lower-ranked choices. Therefore in simulation we evaluate both *decision-level agreement* and *distribution-level fit*, while in real-human evaluation we focus on *decision-level* and *ranking-level* outcomes.

5 Method

5.1 Preference-task generation

Each task is parameterized by a latent preference vector

$$\theta = (\theta^{(1)}, \ldots, \theta^{(m)}), \tag{11}$$

where each coordinate corresponds to a structured trade-off dimension such as cost vs. quality, speed vs. comfort, risk aversion, tool-use preference, conciseness vs. completeness, or proactivity vs. conservatism.

For a given context x, we define a latent utility

$$U_\theta(x,y) = \langle\theta, \phi(x,y)\rangle + \eta(x,y), \tag{12}$$

where $\phi(x,y)$ is a feature map over context-action pairs and η is noise or unobserved nuisance variation. The user decision distribution is

$$P_\theta(y \mid x) = \frac{\exp(U_\theta(x,y)/\tau)}{\sum_{y'\in\mathcal{Y}} \exp(U_\theta(x,y')/\tau)}, \tag{13}$$

or any more general preference generator used by the simulator. This gives a principled latent-variable model for diverse preference structures.

5.2 Adaptive agent

The adaptive agent consists of:

1. a **backbone policy** g_ϕ that proposes decisions;
2. an **interaction encoder** h_ϕ that maps the trajectory $\mathcal{D}_t$ into a latent preference state z_t;
3. an **adaptation rule** that updates z_t online from new behavioral observations.

Concretely,

$$z_t = h_\phi(\mathcal{D}_t), \qquad Q_\phi(y \mid x, \mathcal{D}_t) = g_\phi(y \mid x, z_t). \tag{14}$$

This abstraction covers memory-based adaptation, recurrent latent-state adaptation, in-context preference inference, and gradient-based meta-learning. Our framework is agnostic to the exact implementation and only requires that adaptation be interaction-conditioned.

5.3 Meta-training objective

We sample preference tasks $\theta \sim \Pi_{\text{sim}}$, generate support trajectories $\mathcal{D}_t$ via interactive rollouts with the simulator, and optimize the held-out query loss:

$$\mathcal{L}_{\text{meta}}(\phi) = \mathbb{E}_{\theta \sim \Pi_{\text{sim}}} \left[\mathbb{E}_{\mathcal{D}_t} \mathbb{E}_{(x,y)\sim \mathcal{Q}_\theta} \big[-\log Q_\phi(y \mid x, \mathcal{D}_t) \big] \right], \tag{15}$$

where $\mathcal{Q}_\theta$ is a held-out query set from the same preference task.

When applicable, we augment this with a ranking consistency term:

$$\mathcal{L}_{\text{rank}}(\phi) = \mathbb{E}_{\theta,x} \left[1 - \tau_K\big(\pi_B(x, \mathcal{D}_t), \pi_\theta(x)\big) \right], \tag{16}$$

where τ_K denotes rank agreement over the top-K items. The full training loss is

$$\mathcal{L}(\phi) = \mathcal{L}_{\text{meta}}(\phi) + \lambda \mathcal{L}_{\text{rank}}(\phi). \tag{17}$$

6 Evaluation Metrics

To test whether the agent can fit diverse preference structures, we recommend the following metric suite.

6.1 Decision-level metrics

Exact decision agreement.

$$\text{Acc}_{\text{dec}} = \mathbb{E}_{x\sim\mu_\theta} \left[\mathbf{1}\big(f_B(x, \mathcal{D}_t) = f_\theta(x)\big) \right]. \tag{18}$$

Top-K overlap.

$$\text{TopK} = \mathbb{E}_{x\sim\mu_\theta} \left[\frac{|\text{TopK}(\pi_B(x, \mathcal{D}_t)) \cap \text{TopK}(\pi_\theta(x))|}{K} \right]. \tag{19}$$

Rank correlation. We compute Kendall's τ or Spearman's ρ between the agent ranking and the target ranking.

6.2 Distribution-level metrics (simulation only)

When the target decision distribution is available from the simulator, we additionally report the following distribution-level metrics.

Jensen–Shannon similarity.

$$\text{JSSim} = 1 - \mathbb{E}_{x\sim\mu_\theta} \left[\text{JS}\big(Q_\phi(\cdot \mid x, \mathcal{D}_t), P_\theta(\cdot \mid x)\big) \right]. \tag{20}$$

Cross-entropy fit.

$$\text{CEFit} = -\mathbb{E}_{(x,y)\sim P_\theta} \log Q_\phi(y \mid x, \mathcal{D}_t). \tag{21}$$

6.3 Adaptation-efficiency metrics

Interaction-efficiency curve.

$$\mathrm{AdaptCurve}(t) = \mathbb{E}_\theta \left[\mathrm{Acc}_{\mathrm{dec}}^{(t)}(\theta) \right]. \tag{22}$$

This measures how quickly the agent improves as more interaction data is observed.

Sample complexity to threshold. Let α be a target agreement threshold. Define

$$T_\alpha = \min\{t : \mathrm{Acc}_{\mathrm{dec}}^{(t)} \geq \alpha\}. \tag{23}$$

Smaller T_α indicates more efficient preference adaptation.

6.4 Human-deployment metrics

When the interaction partner is a real human, we evaluate the agent on held-out human choices or rankings over candidate options. We report:

1. **exact decision agreement**;
2. **top-K overlap**;
3. **rank correlation**;
4. **pre/post adaptation improvement**;
5. **human preference fitting gain** over non-adaptive baselines;
6. **user correction cost**, measured by the number of clarifications or manual corrections needed before the agent reaches stable alignment.

We do not compute distribution-level metrics in real-human evaluation.

7 Experimental Design

This paper's empirical claim has two parts:

1. **Mechanism validation:** show that B's interactive learning mechanism can fit diverse preference-induced decision patterns from behavioral data.
2. **Human instantiation:** show that when the interaction partner becomes a real human, the same mechanism can fit the human's future decision behavior.

We therefore propose a two-stage evaluation protocol.

7.1 Stage I: Simulated task-family validation

7.1.1 Goal

To test whether the adaptation mechanism fits diverse preference structures across a broad family of tasks.

7.1.2 Task construction

We instantiate multiple preference dimensions:

- cost vs. quality,
- risk aversion vs. risk seeking,
- speed vs. comfort,
- conciseness vs. completeness,
- proactive assistance vs. minimal intervention,
- tool preference and parameterization style.

Tasks are generated both independently and in compositional combinations, so that the agent must adapt not only to atomic preferences but also to novel mixtures.

7.1.3 Recommended scenarios

We recommend three primary scenarios because they expose explicit decision structures:

1. **Conversational recommendation:** choosing items or rankings under varying latent preferences.
2. **Personalized tool-use agents:** deciding which tool to invoke, with what arguments, under user-specific preferences; this is aligned with the personalized tool invocation setting emphasized by ETAPP [Hao et al., 2025].
3. **Task-oriented personal assistants:** planning and decision support under conflicting user constraints.

7.1.4 Baselines

We compare the proposed method against:

- **Non-adaptive agent:** no interaction-conditioned update.
- **Prompt-only personalization:** user profile given in text but no learned adaptation.
- **Memory-based heuristic adaptation:** raw history concatenation without explicit task-family meta-training.
- **Single-task fine-tuning:** trained on one preference type but not meta-trained over multiple tasks.
- **Oracle profile baseline:** receives the ground-truth preference vector θ.

7.1.5 Main hypotheses

H1: The proposed method achieves higher decision agreement, ranking consistency, and simulated distributional fit than all non-meta-trained baselines.

H2: The proposed method exhibits lower sample complexity T_α and steeper adaptation curves.

H3: The proposed method generalizes better to unseen compositions of preference dimensions.

7.1.6 Key ablations

1. **Preference family coverage.** Vary the diversity and compositional richness of simulated tasks.
2. **Interaction length.** Sweep the number of support rounds t.
3. **Task-family mismatch.** Evaluate how performance changes as test tasks move farther from the meta-training task family.
4. **Adaptation architecture.** Compare recurrent adaptation, in-context adaptation, latent-state update, and gradient-based inner-loop updates.

7.1.7 What would count as proof of mechanism validity?

We cannot prove a universal property empirically, but we can provide strong evidence. The mechanism is considered validated if:

1. the agent consistently improves with more interaction rounds across diverse task types;
2. the improvement transfers to unseen preference compositions;
3. gains are largest precisely in metrics that diagnose preference fitting rather than general task competence;
4. the learned adaptation mechanism outperforms non-adaptive and narrowly tuned baselines.

7.2 Stage II: Direct real-human preference fitting

7.2.1 Goal

To test the actual deployment claim: once the interaction partner is replaced by a real human, can the same adaptation mechanism fit the human's future decision behavior?

7.2.2 Protocol

For each human participant:

1. Sample an initial set of interaction contexts.
2. For each context, present a fixed candidate set of decisions or responses.
3. Let the human interact freely with the adaptive agent over t rounds, providing choices or rankings over the candidate options.
4. Hold out a future set of contexts not seen during adaptation.
5. Compare the agent's post-adaptation decisions or rankings against the human's held-out choices or rankings.

7.2.3 Measures

We report:

- exact decision agreement on held-out contexts,
- top-K overlap,
- rank correlation,
- interaction efficiency,
- user correction cost,
- pre/post adaptation gain.

We do not report distribution-level metrics in Stage II, because real-human evaluation elicits choices or rankings over candidate options rather than a full decision distribution.

7.2.4 Main hypotheses

H4: The same adaptation mechanism that succeeds in Stage I yields significant pre/post improvement in fitting held-out human choices or rankings.

H5: Simulated-task adaptation performance predicts human adaptation performance when the human domain is well aligned with the simulated task family.

7.2.5 Human-study design considerations

- Balance preference types across participants.
- Include repeated and novel contexts to disentangle memorization from preference inference.
- Use fixed candidate sets when ranking-based evaluation is required.
- Collect demographic and behavioral metadata only where ethically appropriate and necessary.
- Pre-register exclusion criteria and maintain transparency about participant compensation and consent.

7.3 Stage III: Stress tests and falsification

A strong paper should include experiments that could falsify the thesis under violated assumptions.

Stress Test 1: Out-of-family human preferences. Recruit users whose preferences depend on dimensions absent from the simulated family. If performance drops sharply, this supports the task-family alignment requirement.

Stress Test 2: Large task mismatch. Construct evaluation tasks that are progressively farther from the simulated task family under d_{Θ}. The theorem predicts degradation as the mismatch term grows.

Stress Test 3: Behavioral non-stationarity. Allow human preferences to shift over time. This tests whether the adaptation mechanism can track evolving preference patterns rather than static profiles.

Stress Test 4: Simulator-only overfitting. Compare an agent optimized solely for simulator agreement against the proposed meta-learned adaptation agent. The former may overfit simulator artifacts; the latter should better fit decision-relevant preference patterns and thus transfer more reliably to humans.

8 Expected Results and Statistical Analysis

8.1 Primary expected findings

We expect the following pattern:

1. Meta-trained interactive adaptation outperforms non-adaptive and prompt-only baselines on simulated task-family preference fitting.
2. Gains are especially pronounced in low-data regimes, consistent with the meta-learning hypothesis.
3. The same adaptation mechanism yields significant improvement in direct human preference fitting, measured on held-out human choices or rankings.
4. The predictive power of simulated-task performance is strongest when the simulated preference family is well aligned with the structure of the human task.

8.2 Statistical testing

For all metrics, we recommend:

- reporting mean $\pm$ standard error,
- bootstrap confidence intervals,
- paired permutation tests for pre/post adaptation changes,
- mixed-effects models when analyzing participant-level variation,
- correlation analysis between Stage I and Stage II metrics.

A key analysis is the correlation between simulated adaptation success and human fitting success:

$$\rho = \mathrm{corr}\left(\mathrm{StageI_AdaptScore}, \mathrm{StageII_HumanFitScore}\right). \tag{24}$$

A positive and significant ρ supports the claim that simulation validates the adaptation mechanism, not the simulator itself.

9 Discussion

What this paper does *not* claim. We do not claim that high agreement with a simulator implies high agreement with real humans. Recent evidence strongly warns against such a conclusion [Seshadri et al., 2026, Zhou et al., 2026]. We also do not claim exact identification of a user's latent preference parameter from finite interaction. Our thesis is narrower: simulation is used to validate a *learning capability* over a task family.

What this paper *does* claim. If the adaptive mechanism can fit decision-relevant preference patterns across a broad and controlled family of tasks, then it is reasonable to expect the same mechanism to fit real human preferences *through direct human interaction*, provided that the human task family is sufficiently close to the simulated task family.

Meta-learning as the right abstraction. The central methodological benefit of the meta-learning view is that it forces the evaluation to target rapid adaptation rather than static personalization. This is crucial for personalized agents, because the deployment-time problem is almost always few-shot: the agent must infer user preferences from limited behavioral evidence.

10 Limitations

Our framework depends on approximate alignment between simulated task families and human preference dynamics. If humans exhibit preference dimensions not captured in simulation, transfer can fail. The theory also assumes that nearby tasks induce nearby fitting risk for the learned adaptation mechanism; this may fail under highly discontinuous behavior or brittle model updates. Finally, some important forms of preference are non-stationary or reflexive: humans may revise their own preferences during interaction. Extending the framework to such settings is an important direction for future work.

11 Conclusion

This paper proposes a principled answer to a fundamental question in personalized agent research: what can simulation actually validate? We argue that simulation should validate not the simulator as a human proxy, but the agent's interactive learning mechanism as a fitter of decision-relevant preference patterns. Motivated by meta-learning, we formulate preference adaptation as a task-family generalization problem, derive a transfer guarantee under approximate task-family alignment and risk smoothness, and design a complete evaluation protocol that separates mechanism validation from human deployment. The resulting thesis is both mathematically precise and empirically testable: if the agent can learn to fit diverse preference-induced decision behavior in simulation, then under small task-family mismatch, it should be able to fit human future decisions when interacting with real humans directly.

References

Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017. Available at `https://arxiv.org/abs/1703.03400`.

Yizhong Hao and colleagues. Evaluating personalized tool-augmented LLMs from the perspective of agents. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. Available at `https://aclanthology.org/2025.acl-long.1064/`.

Preethi Seshadri, Samuel Cahyawijaya, Ayomide Odumakinde, Sameer Singh, and Seraphina Goldfarb-Tarrant. Lost in simulation: LLM-simulated users are unreliable proxies for human users in agentic evaluations. *arXiv preprint arXiv:2601.17087*, 2026. Available at `https://arxiv.org/abs/2601.17087`.

Xuhui Zhou, Weiwei Sun, Qianou Ma, Yiqing Xie, Jiarui Liu, Weihua Du, Sean Welleck, Yiming Yang, Graham Neubig, Sherry Tongshuang Wu, and Maarten Sap. Mind the Sim2Real gap in user simulation for agentic tasks. *arXiv preprint arXiv:2603.11245*, 2026. Available at `https://arxiv.org/abs/2603.11245`.